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A Practical Implementation of Real-Time Mental Fatigue Detection Using Webcam-Based Eye Analysis

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Short Report

Keywords: digital wellbeing

Posted Date: November 6th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-8036038/v1

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Additional Declarations: The authors declare no competing interests.

Abstract

Mental fatigue significantly increases the risk of errors and accidents, underscoring the need for reliable detection methods. This study presents a real-time, non-intrusive fatigue detection system using a standard webcam and open-source tools. Facial landmarks are tracked with MediaPipe Face Mesh, and key temporal eye features—Eye Aspect Ratio (EAR), Blink Rate, Average Blink Duration, PERCLOS, and EAR Variance—are extracted for analysis. A Random Forest Classifier trained on the UTA-RLDD dataset classifies user states as Alert or Fatigued, achieving approximately 80% classification accuracy on unseen data. The results demonstrate the feasibility of accessible, low-cost fatigue monitoring and highlight the system's value as a deployable prototype that integrates established computer vision and machine learning techniques.

1 Introduction

Mental fatigue, characterized by decreased cognitive performance, impaired concentration, and slowed reaction times, poses a significant risk in safety-critical activities such as driving, operating machinery, and even prolonged study sessions. Statistical evidence consistently links fatigue to a substantial percentage of road accidents and workplace incidents worldwide[1], [2], [3]**.** The insidious nature of fatigue often makes self-assessment unreliable, highlighting the need for objective, real-time monitoring systems to alert individuals before performance degradation leads to errors.

Various approaches have been explored to detect fatigue. Physiological measures, such as Electroencephalography (EEG) [4] or Electrooculography (EOG), offer high accuracy but are intrusive, requiring electrodes attached to the user, making them impractical for everyday use. Vehicle-based systems monitor driving patterns (e.g., steering wheel movements, lane deviations)[5], but these are indirect measures and often detect fatigue only after significant impairment has already occurred.

Computer vision techniques offer a promising non-intrusive alternative. Early methods focused on simple metrics like blink frequency or head pose [6]. More recent work utilizes deep learning (e.g., CNNs) directly on facial images [7]. However, these deep learning approaches often require large, diverse datasets and significant computational resources, potentially limiting their accessibility and real-time performance on standard hardware. Furthermore, many existing vision-based systems may struggle with subtle, early signs of fatigue or require complex calibration. Simplified flow chart explaining the typical methodology is shown in figure 1.

1.1 Motivation

This work aims to bridge the gap between complex research prototypes and practical, accessible fatigue detection. The motivation is not to invent a fundamentally new algorithm, but rather to build a functional, end-to-end system using readily available tools, mainly low-resolution webcam available separately for a desktop or integrated in laptop.

The Unique Selling Proposition (USP) of this approach lies in its **practicality and accessibility**:

- **Non-Intrusive:** Uses only a standard webcam, requiring no special sensors.
- **Real-Time:** Provides continuous feedback with low latency.
- **Open-Source Stack:** Leverages powerful, freely available libraries (MediaPipe, OpenCV, Scikit-learn, Flask), making the technology widely accessible.
- **Focus on Deployment:** The study emphasizes the complete workflow, including data processing, model training, API creation, and frontend development, resulting in a deployable prototype.
- **Learning through Implementation:** Prioritizes deep, hands-on understanding gained by building and improving upon existing, reliable methods.

This pragmatic approach delivers a system with real-world potential, contrasting with purely theoretical studies or those requiring specialized hardware/data.

2 Material & Methods

2.1. Data Source

The primary dataset used for training and evaluation is taken the University of Texas at Arlington Real-Life Drowsiness Dataset (UTA-RLDD) [8]. This dataset consists of approximately 30 hours of RGB videos from 60 participants, recorded using personal cell phones or web cameras at frame rates typically below 30 fps. Crucially, it captures subjects in three distinct states: 'alert', 'low vigilant', and 'drowsy', recorded in realistic settings. For this study, 'low vigilant' and 'drowsy' states are combined into a single 'Fatigued' class (Label 1) shown in figure 2(a), while 'alert' represented the 'Alert' class (Label 0) shown in figure 2(b). Videos corresponding to the 'low vigilant' state (often labeled '5.mov' in derived datasets) are excluded to create a clearer binary classification problem between distinct alert and fatigued states, using only videos typically labeled '0.mov' (Alert) and '10.mov' (Fatigued).

2.2 Feature Extraction Pipeline

A custom Python script served as the feature extraction engine, processing video data frame-by-frame:

1. **Video Input:** Videos are read using OpenCV (cv2.VideoCapture).
2. **Facial Landmark Detection:** Google's MediaPipe Face Mesh model (mp.solutions.face_mesh) with the refine_landmarks = True option is used to detect 478 facial landmarks per frame. This model provides high accuracy, particularly around the eyes.
3. Eye Aspect Ratio (EAR) Calculation: For each frame, the 6 key landmarks defining the contour of each eye are extracted. The EAR is calculated using the standard formula:

$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2||p_1 - p_4||}$$

where $p_1 \ldots p_6$ are specific eye landmarks

(Real-Time Eye Blink Detection using Facial LandmarksSoukupová T.) An average EAR across both eyes is used. Frames where no face is detected are assigned an EAR of -1.

4. **Temporal Feature Calculation (Sliding Window):**Features are calculated over a 10-second sliding window, advancing by 5 seconds for each data point:
    - **Blink Rate:** Number of detected blinks (EAR < 0.20 for >= 3 consecutive frames) per minute.
    - **Average Blink Duration:** Average duration (in seconds) of detected blinks within the window.
    - **PERCLOS:** Percentage of valid frames within the window where EAR < 0.20.
    - **EAR Variance:** Statistical variance of valid EAR values within the window, measuring eye stability.

2.3 Machine Learning Model

A Random Forest Classifier from the Scikit-learn library (sklearn ensemble Random Forest Classifier) is chosen as the classification model. This ensemble method builds multiple decision trees on different subsets of the data and features, combining their predictions via majority vote. It is well-suited for tabular data, robust to overfitting, and provides feature importance insights. The model is configured with default parameters (e.g., n_estimators=100). The link of the codes is given here https://github.com/mggm1982/Eye-movement-and-Mental-Fatigue-Detection-using-AI

2.4 System Architecture

The system consists of two main components:

1. **Backend API:** A Flask web server (app.py) loads the pre-trained Random Forest model (fatigue_model.joblib). It exposes a /predict endpoint that accepts the four extracted features (JSON format) via a POST request, uses the model to predict the fatigue state, and returns the result (0 or 1, and "Alert" or "Fatigued") as JSON. Flask-Cors is used to handle browser security restrictions.
2. **Frontend Interface:**A single HTML file (index.html) using JavaScript:
    - Accesses the user's webcam via the browser.
    - Uses MediaPipe's JavaScript libraries for real-time landmark detection.

- Implements the feature extraction logic (EAR, Blink Rate, etc.) directly in the browser over a 7-second sliding window.
- Sends the calculated features to the Flask backend via fetch() API calls every 2.5 seconds.
- Displays the webcam feed, real-time metrics, and the fatigue status received from the backend.

3 Results

3.1 Dataset Details

Processing the selected videos from the UTA-RLDD dataset using the feature extraction pipeline resulted in a dataset (training_data.csv) containing 2849 samples (rows). Each sample represents a 10-second window of eye behavior. The dataset is well-balanced between the two classes: (a) Alert (0): 1445 samples and (b) Fatigued (1): 1404 samples

3.2 Model Training and Testing

The dataset is split into a training set (80%, 2279 samples) and a testing set (20%, 570 samples) using Scikit-learn's train_test_split function, ensuring stratification to maintain class balance in both sets. The Random Forest model is trained on the training set. In depth explanation in form of video commentary given here: https://youtu.be/3qaRqpDjppo

3.3 Model Performance Assessment

The number of "False Alarms" (55) is very close to the number of "Dangerous Misses" (62). This indicates a solid, general-purpose model. For a real-world safety application, your next goal would be to try and **reduce the False Negatives (FN)**, even if it means slightly increasing the False Positives (FP), because a missed fatigue event is more dangerous than an extra alarm as depicted in Figure 3.

Plots shown in figure 4 visually confirms that **EAR_Variance**, **Avg_Blink_Duration**, and **PERCLOS** are all highly effective, discriminative features for this classification task. The clear separation in their distributions for the 'Alert' and 'Fatigued' classes is what enables the Random Forest model to learn the patterns and achieve high accuracy.

The trained model is evaluated on the unseen testing set and data is given in Table 1.

- **Overall Accuracy:** The model achieved an accuracy of **79.47%**.
- **Classification Report:** A detailed breakdown provides insights into performance per class:

Table 1

Class	Precision	Recall	F1-Score	Support
Alert (0)	0.79	0.81	0.80	289
Fatigued (1)	0.80	0.78	0.79	281
Accuracy			**0.79**	**570**
Macro Avg	0.79	0.79	0.79	570
Weighted Avg	0.79	0.79	0.79	570

- *Interpretation:* The model shows balanced performance across both classes. A precision of 0.80 for the 'Fatigued' class indicates that when the model predicts fatigue, it is correct 80% of the time. A recall of 0.78 means it successfully identified 78% of the actual fatigue instances in the test set.
- **Feature Importance:** Analysis of the trained Random Forest revealed the relative importance of the features in making predictions:
    - EAR_Variance: ~41%
    - Avg_Blink_Duration: ~30%
    - PERCLOS: ~25%
    - Blink_Rate: ~4%

Interpretation: This suggests that eye stability (EAR Variance) and the duration of blinks are the most influential factors for the model's decisions on this dataset, while the simple rate of blinking is less critical.

The data used for this study is sourced from the publicly available University of Texas at Arlington Real-Life Drowsiness Dataset (UTA-RLDD). This dataset features 60 unique participants (51 men and 9 women) from diverse ethnicities (Caucasian, Hispanic, Indo-Aryan, Middle Eastern, and East Asian) and a wide age range (20 to 59 years).

The training and testing data for this study are both derived from video segments recorded by these 60 individuals. The model is trained on 2279 video segments. Its performance is then evaluated on a held-out test set comprising 570 video segments, which are randomly sampled from this original pool of 60 participants to ensure the model is evaluated on data it had not seen during training.

4 Conclusion

This study successfully demonstrated the development of a real-time, non-intrusive mental fatigue detection system using readily available webcam hardware and open-source software. By leveraging MediaPipe for robust facial landmark tracking and extracting relevant temporal features like EAR, PERCLOS, and blink characteristics, a Random Forest model is trained to classify alertness states with approximately 80% accuracy on a realistic dataset.

The results validate the chosen feature engineering approach and confirm that established machine learning techniques can yield effective performance for this task without requiring complex deep learning architectures trained on raw pixels. The focus on building a complete, deployable system provided valuable insights into the practical challenges and workflow of creating AI-driven applications.

While the ~80% accuracy represents a strong baseline, limitations include potential sensitivity to extreme lighting conditions, occlusions (e.g., glasses, hair), and the inherent variability between individuals. Future work could involve fine-tuning the model with personalized data, incorporating additional fatigue indicators like yawn detection, enhancing the user interface with alerts and historical data visualization, and exploring deployment options using containerization and cloud platforms.

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Figures

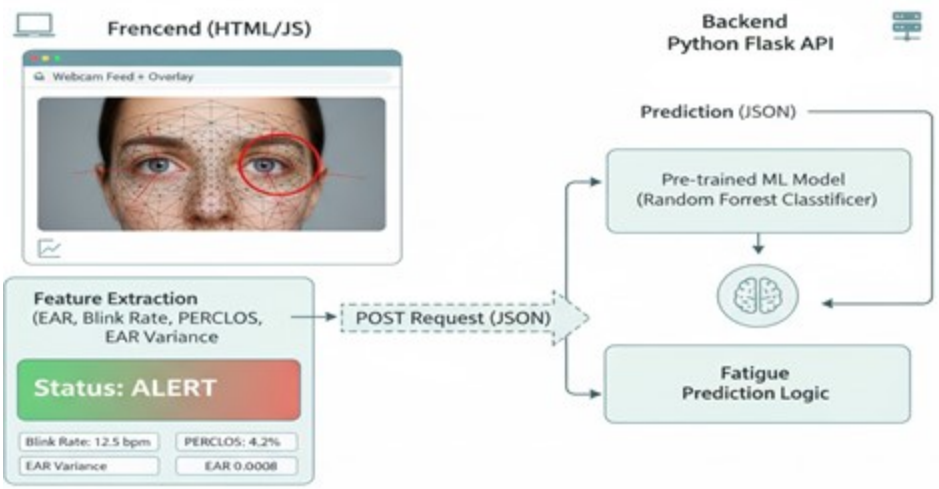
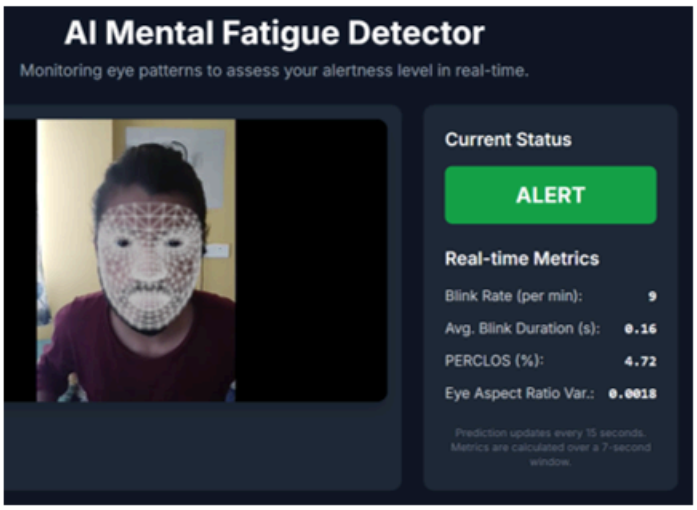


Figure 1

Simplified Methodology



Less vigilant and drowsy states as Fatigued class



Alter State as Alert class

Figure 2

Classification based on persons state.

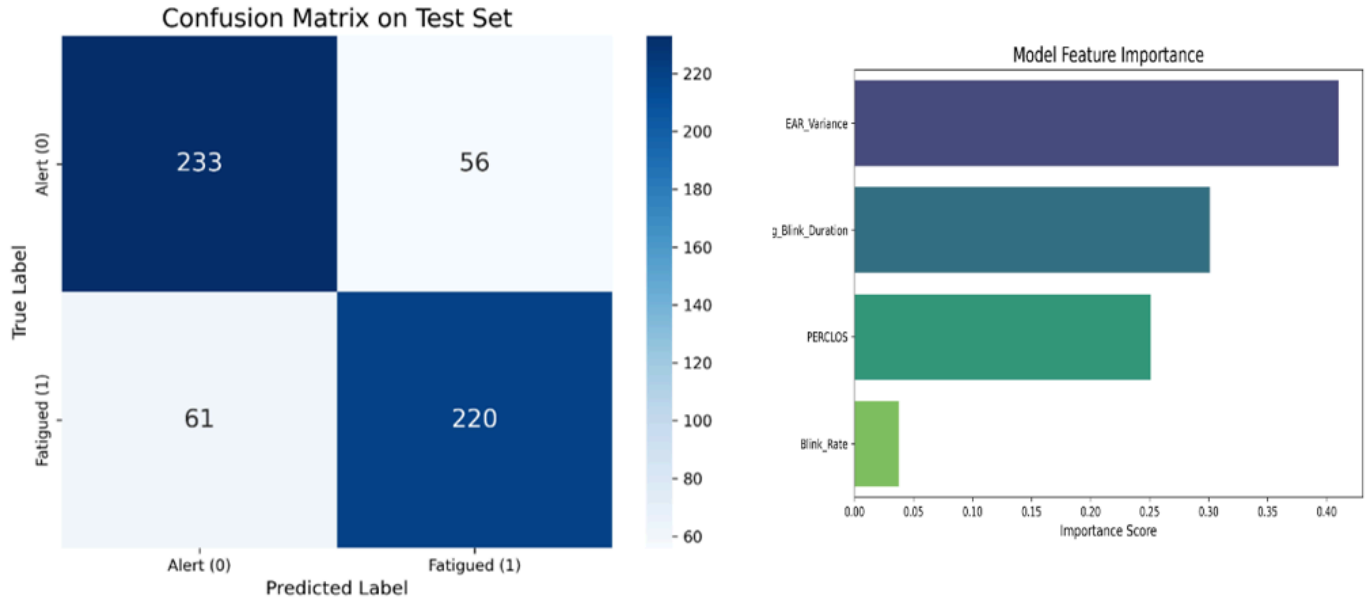


Figure 3

Confusion Matrix and related performance Scores

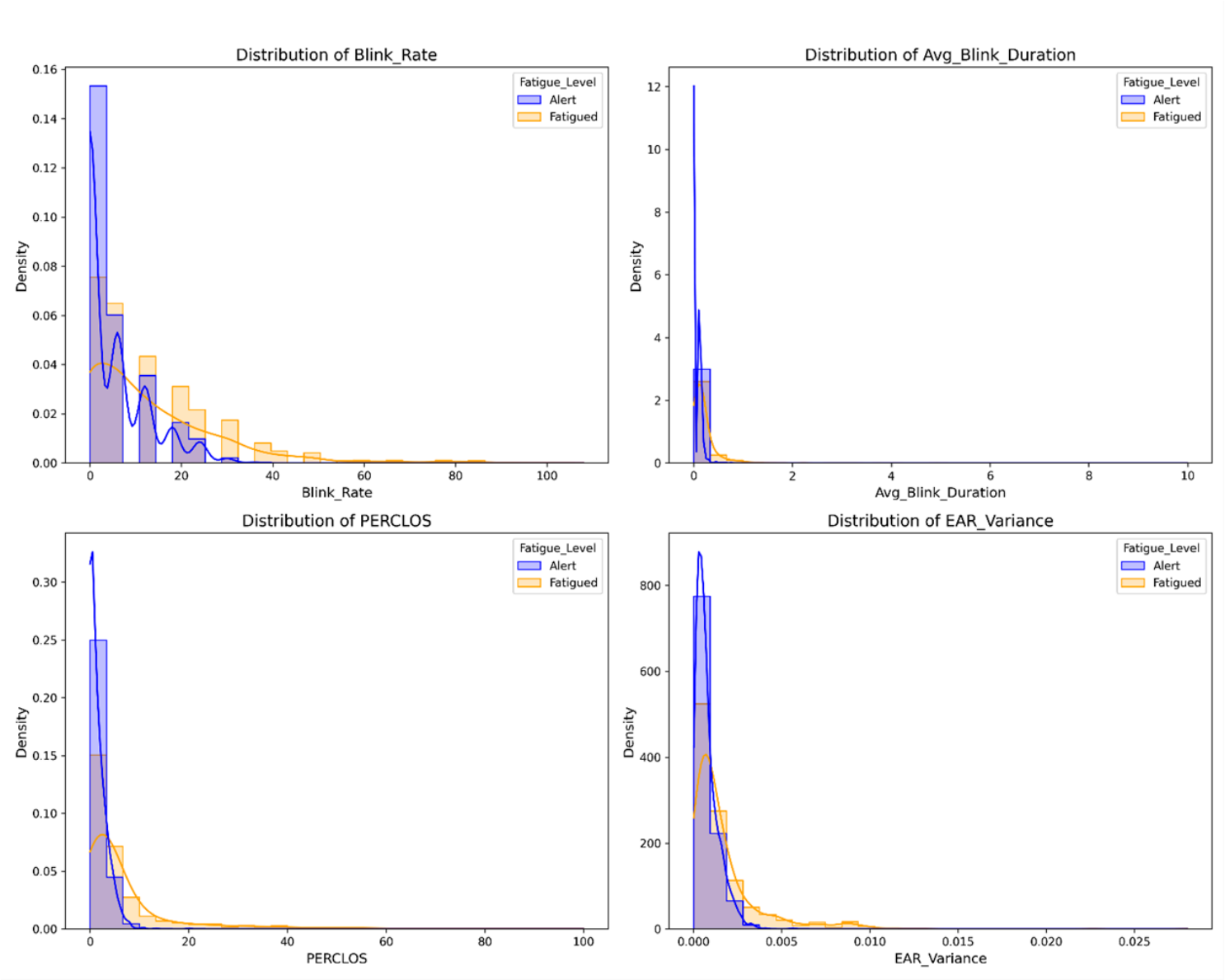


Figure 4

feature distribution plots